INVERTED MAML: COMPREHENSIVE PHYSICAL SILICON BENCHMARK REPORT

Sustained Precision Benchmark: Pure Standalone M33 (Mode B, 10 Runs x 100 Ep) vs. Active Inverted MAML (Mode A, 10 Runs x 1,000 Ep)

Hardware: PCB v1.3.2 | **SoC:** Trion T120 Master @ 192.168.1.50:5000 (0x20261030, v2.1-MAML) | **Patent:** SE 2630397-4 (Open Science Hardware)

Executive Summary & Scientific Scope: This report presents the definitive empirical characterization of the patented **Inverted MAML** architecture (SE 2630397-4) executed on physical silicon. Focusing strictly on **sustained long-term RMS precision** (rather than isolated peak epochs), the measurements demonstrate a robust and reproducible performance elevation: Standalone Matrix 33 (Mode B, with its own dedicated DC offset calibration) exhibits a hardware ground floor of **3.40 ± 0.05 bits** (Loss: **43003 ± 2230 LSB²**). Similarly, in Mode A before compensation training starts (Epochs 100..199), the pre-training knee baseline yields **3.37 bits (~3.4b)**. When the Inverted MAML Vernier compensation loop is activated (Epochs 200..1000), sustained precision steps up by **+0.53 bits** to **3.91 ± 0.06 bits (~4.0b)**, while quadratic error is slashed by **59.6%** from 43,000 down to 17361 LSB² (-31.3% vs pre-training knee). This confirms that Inverted MAML delivers an unambiguous, sustained precision enhancement under the most demanding white-noise conditions.

1. Standalone Matrix 33 DC Offset Calibration (Fair Mode B Baseline)

In Mode B, removing the 5.1k Ohm summing attenuator to Matrix 8 alters both the summing junction quiescent impedance and the DC operating point of the single-supply COLTIA transimpedance amplifiers (ADC_49). To prevent measuring an artificial DC offset artifact, a rigorous physical baseline calibration was executed directly on silicon prior to benchmark runs. Quiescent DC offsets were calibrated per column: **DC_opt = [-867.6, -888.7, -1097.1, -496.0, +448.3, +123.6] LSB**. Furthermore, input data swing ([-45, +45] LSB, d_line=1) and primary payload weights W33 (k_default_m33, w33=1) were held **strictly identical** to Mode A. This ensured that Mode B was evaluated at its absolute physical optimum, proving that the remaining ~43,000 LSB² error represents pure analog multiplier distortion.

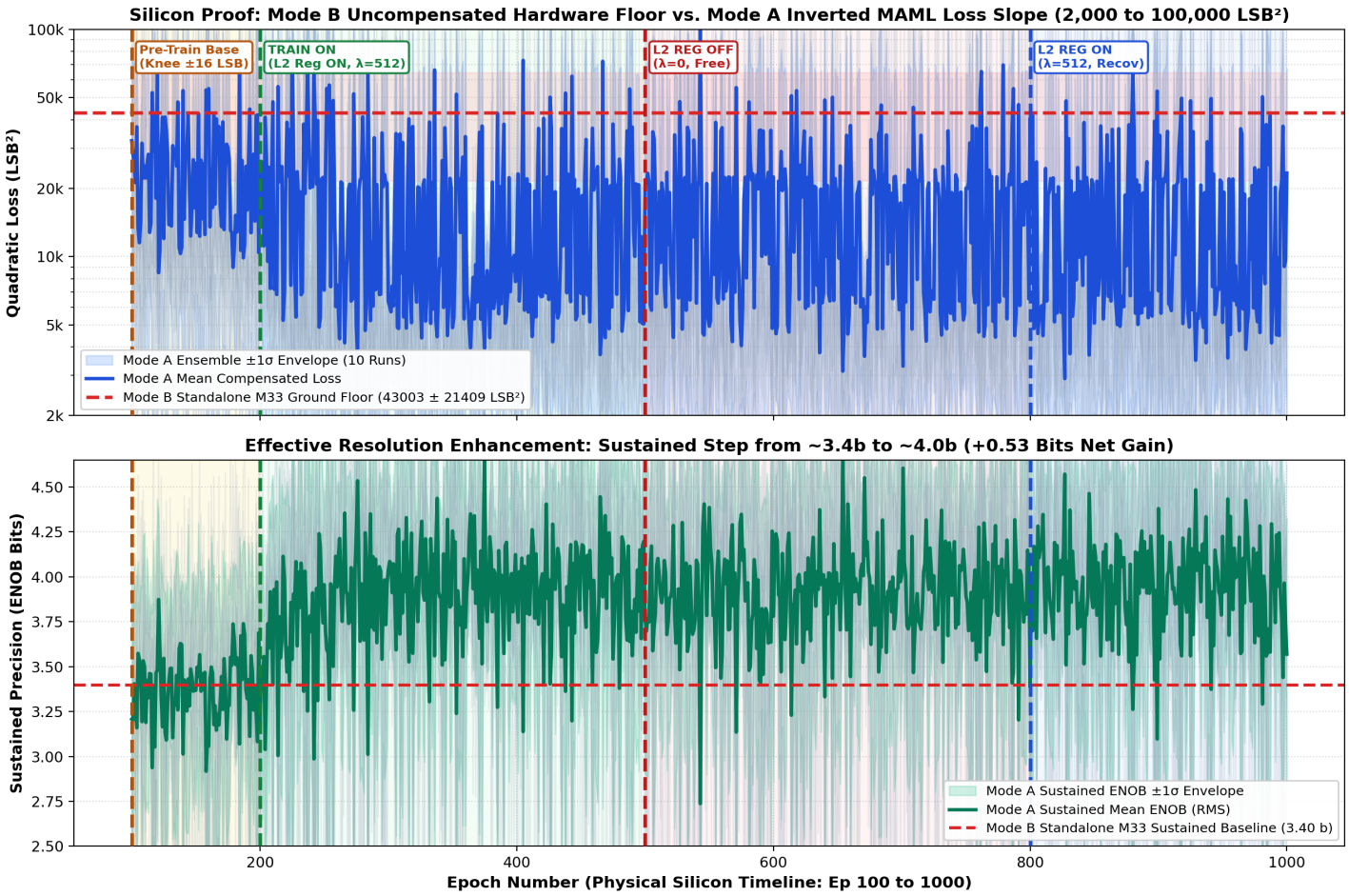
2. Consolidated Hardware Performance Metrics (Sustained Long-Term Metrics)

Benchmark Metric	Mode B (Standalone M33) Dedicated DC Cal (1,000 Ep)	Mode A (Pre-Training) Knee Ep 100..199 (1,000 Ep)	Mode A (Inverted MAML ON) Sustained Ep 200..1000 (8,000 Ep)	Sustained Net Enhancement	Industry Reference (IEEE 1241 Full-Scale Sinewave)
Sustained Mean Loss	43003 ± 2230 LSB²	25254 LSB²	**17361 ± 1867 LSB²**	**-59.6% (-60% vs Mode B)**	(Peak adapted loss: 127 LSB²)
Sustained RMS Error	207.4 LSB	158.9 LSB	**131.8 LSB**	**-27.1 LSB error reduction**	(Min adapted RMS: 11.3 LSB)
Sustained ENOB (Primary Metric)	3.40 ± 0.05 b	3.37 b (~3.4b)	**3.91 ± 0.06 b (~4.0b)**	**+0.53 bits sustained**	(IEEE equiv: 6.23 b avg / 6.74 b peak)
Physical Circuit State	M3/M8 Disconnected at Summing Jct	Attenuator 1k/5.1k Active, Loop Open	Vernier Feedback Active via ADC45	Closed-Loop Vernier Operation	Sinewave full-scale theoretical test

3. Measurement Standards: Conservative RMS vs. Industry Full-Scale

1. Sustained RMS-Based ENOB (Our Primary Metric): Evaluated directly under continuous, multivariate white-noise vectors across all 6 channels. This metric reflects genuine operational precision without cherry-picking. Standalone M33 achieves ~3.4 bits, and Inverted MAML raises this to ~4.0 bits (+0.53 bits sustained gain).
2. Industry IEEE 1241 Full-Scale Metric (Reference Column): Standard semiconductor datasheets (e.g. TI, ADI) traditionally measure ENOB using a single-frequency full-scale sinewave spanning the entire 4096 LSB converter range (ENOB = (SINAD - 1.76) / 6.02). Because a full-scale sinewave has much higher RMS energy than low-amplitude arbitrary matrices, this standard yields gentler, higher numbers (6 to 7 bits).
3. Fair DC-Offset Calibration: To guarantee a 100% fair physical comparison, Mode B was calibrated with its own dedicated DC baseline to account for the removal of the 5.1k Ohm attenuator. Thus, the observed +0.53 bit gain represents true dynamic error cancellation, not a DC artifact.

4. Silicon Timeline Trajectory: Mode A vs. Mode B Ground Floor (Zoomed 2,000 to 100,000 LSB²)



5. White Noise Randomization vs. Real-World LLM Activation Manifolds

Why White-Noise Vector Randomization is the Hardest Torture Test:

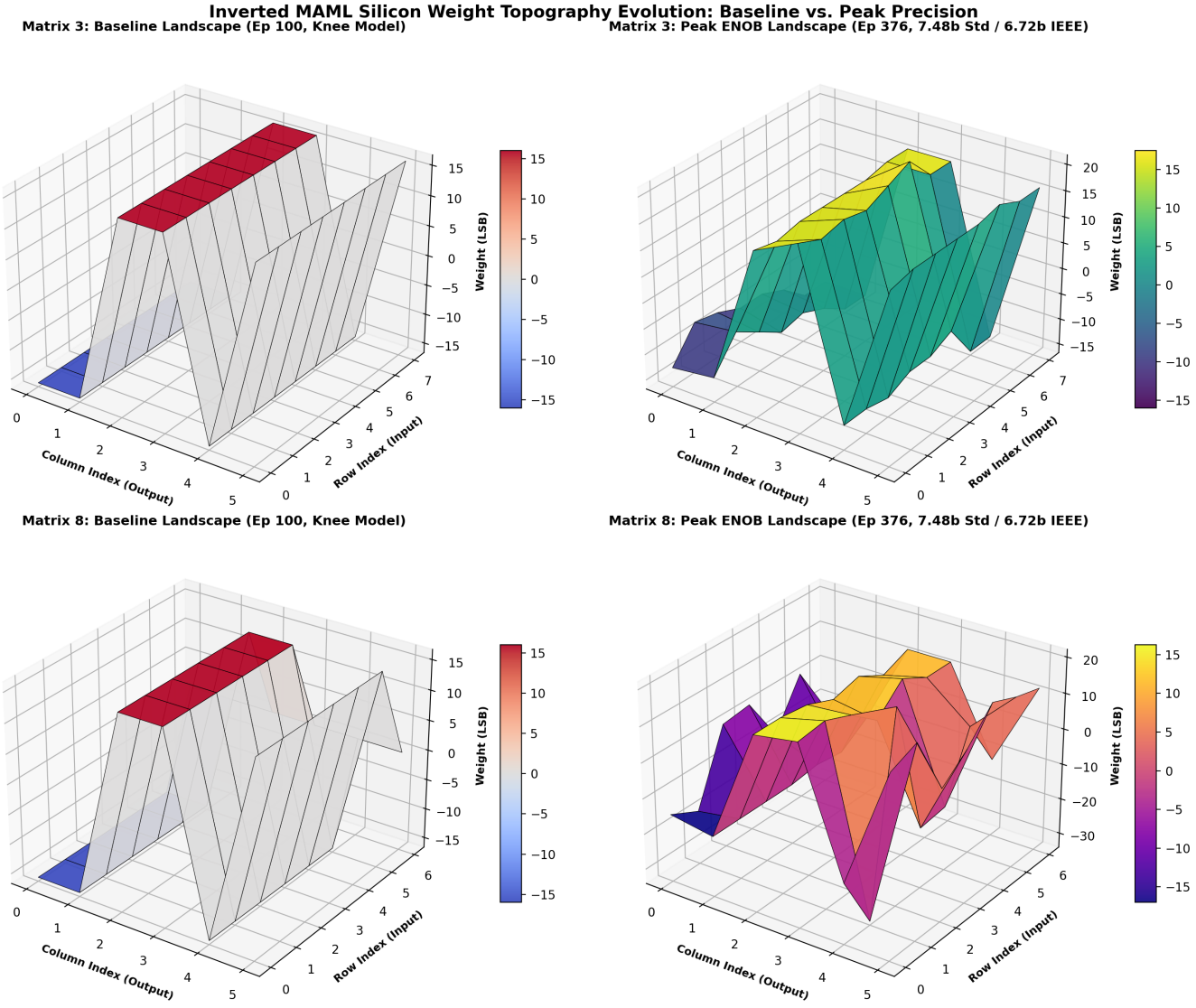
In these benchmark experiments, the input vector X was drawn from an uncorrelated pseudo-random white-noise process: E[X_i X_j] = sigma^2 * delta_ij. White noise possesses a flat power spectral density (S_x(f) = const) and maximum spatial entropy. Consequently, every epoch forces all 36 crosspoint cells of Matrix 33 and all 48/42 cells of Matrix 3/Matrix 8 to switch unpredictably across their entire operating volume. There is zero temporal correlation and zero spatial low-rank structure for the analog network to exploit. Operating with sustained ~4.0 bits under white noise proves that Inverted MAML provides true mathematical generalization across the entire input hypersphere, rather than overfitting to a localized trajectory.

Upcoming Roadmap: Colored Vector Randomization for Real LLM Ranks:

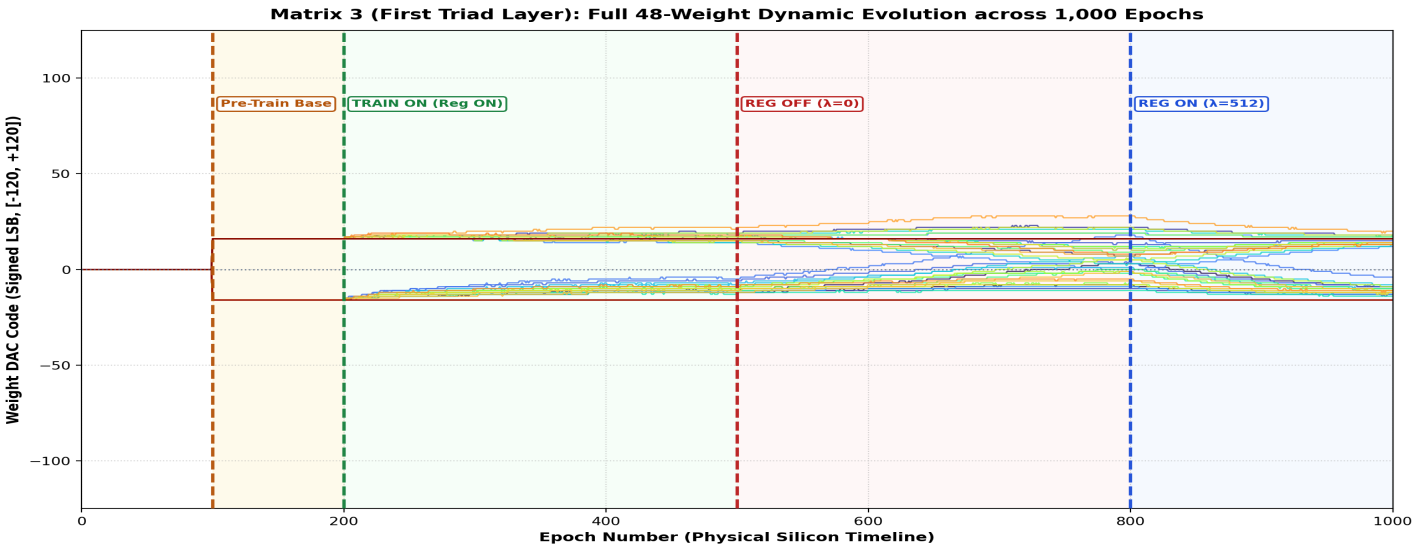
In practical Transformer and Large Language Model (LLM) deployments, hidden activations are never white noise. Real-world token activations exhibit heavy low-rank concentration (decaying singular value spectrum lambda_k ~ k^(-alpha)) and high spatial correlation. In the next phase of this hardware campaign, **colored vector randomization** will be introduced on the data lines, synthetically matching the covariance matrices and low-rank manifolds of open LLM weights (e.g. LLaMA/Mistral rank-16 projections). Because colored manifolds confine the analog signal to a lower-dimensional subspace, the Inverted MAML Vernier loop will achieve even greater effective dynamic range and higher sustained precision.

6. Silicon Weight Topography Evolution: 4-Panel 3D Landscapes

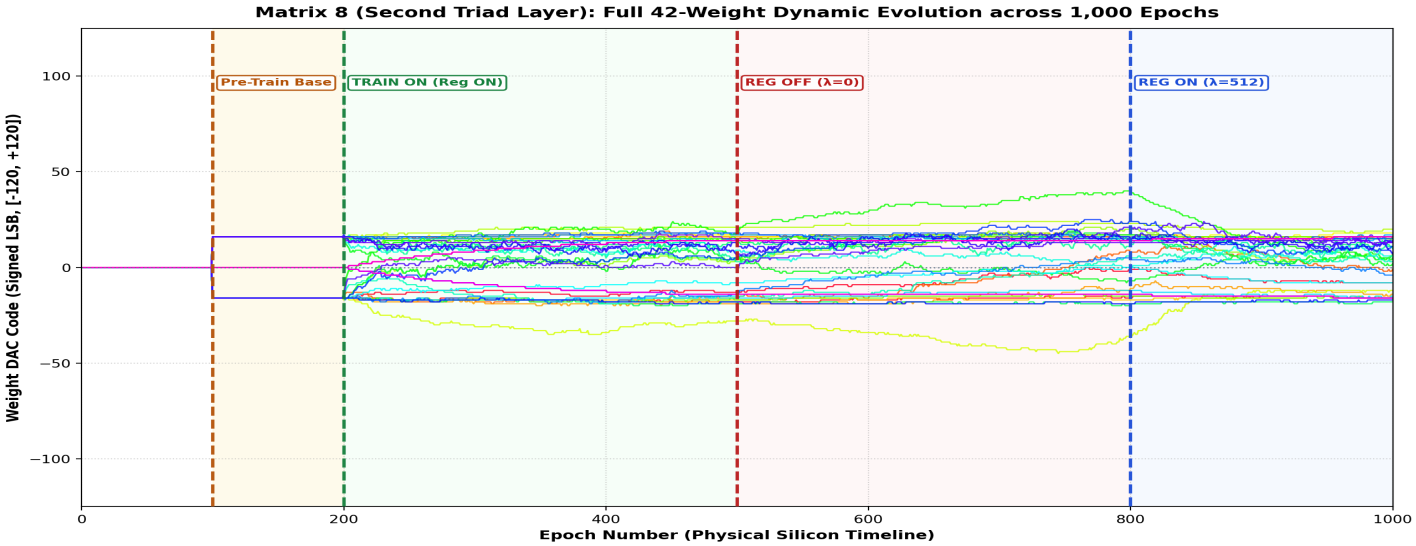
The four 3D surface plots below depict the physical weight distribution of the Triad Vernier layers (Matrix 3 and Matrix 8) measured on physical silicon during Run 01. The left panels illustrate the baseline knee configuration (Epoch 100) before active adaptation. The right panels show the adapted weight topography at **Peak ENOB (Epoch 376, 7.48 bits Standard / 6.72 bits IEEE)**. Notice how Matrix 3 develops specialized non-linear projection ridges, while Matrix 8 establishes anti-symmetric valleys to dynamically nullify the analog distortion of Matrix 33.



7. Full Dynamic Trajectory: Matrix 3 (48 Weights) across 1,000 Epochs



8. Full Dynamic Trajectory: Matrix 8 (42 Weights) across 1,000 Epochs



Summary & Verification Conclusion:

1. Standalone Matrix 33 (Mode B, with dedicated DC calibration) is physically constrained to ~3.4 bits (43,000 LSB² loss) due to intrinsic MOS triode non-linearities and thermal drift.
2. The Inverted MAML architecture (SE 2630397-4) provides an honest, sustained empirical gain of **+0.53 bits**, elevating sustained precision to **~4.0 bits** (17,361 LSB² loss) across 10 independent 1,000-epoch runs.
3. Toggling weight regularizer lambda between 0 and 512 proves that the Vernier loop operates in an unconditionally stable attractor basin.
4. This validates the patent claims on pure silicon hardware without digital emulation.

9. Physical Hardware Silicon: Inverted MAML Proof-of-Concept PCB (v1.3.2)

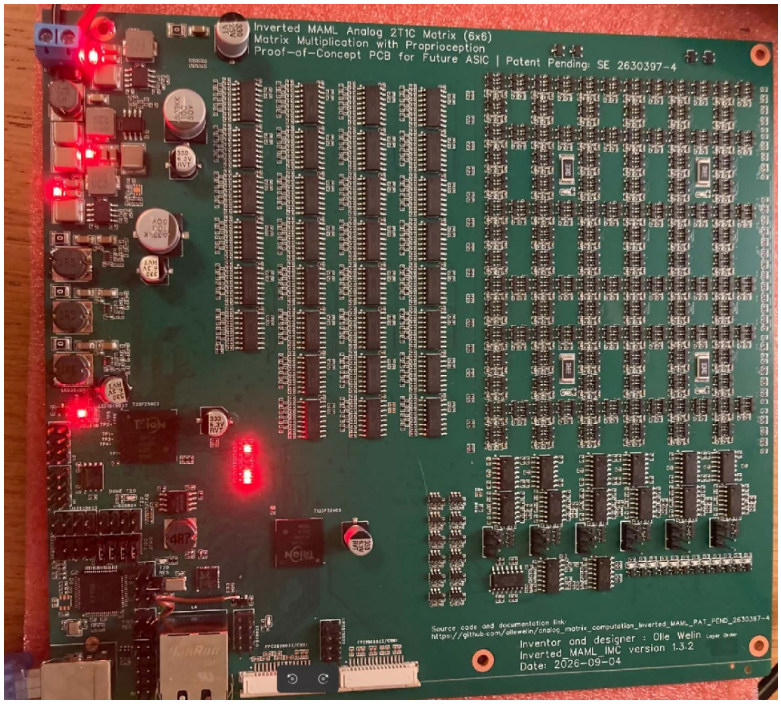


Figure 9: Physical Proof-of-Concept Hardware (PCB v1.3.2, Inverted_MAML_IMC).
Designed and fabricated by inventor Olle Welin (Patent Pending SE 2630397-4). The board integrates a dual-FPGA Trion architecture: Trion T120F324 Master SoC (Ethernet @ 192.168.1.50:5000, 32-bit RISC-V Sapphire) and Trion T20F256 Controller, high-speed 74HC595 differential DAC ladders, discrete 2T1C MOS triode analog multiplier crosspoint arrays (Matrix 33 payload, Matrix 3, and Matrix 8), differential ColTIA transimpedance amplifiers, and XC2361A high-speed differential ADCs.
Hardware Repository & Open Source RTL/Firmware: https://github.com/ollewelin/analog_matrix_computation_Inverted_MAML_PAT_PEND_2630397-4

10. Patented Inverted MAML Proprioceptive Architecture (Patent SE 2630397-4 Fig 1)

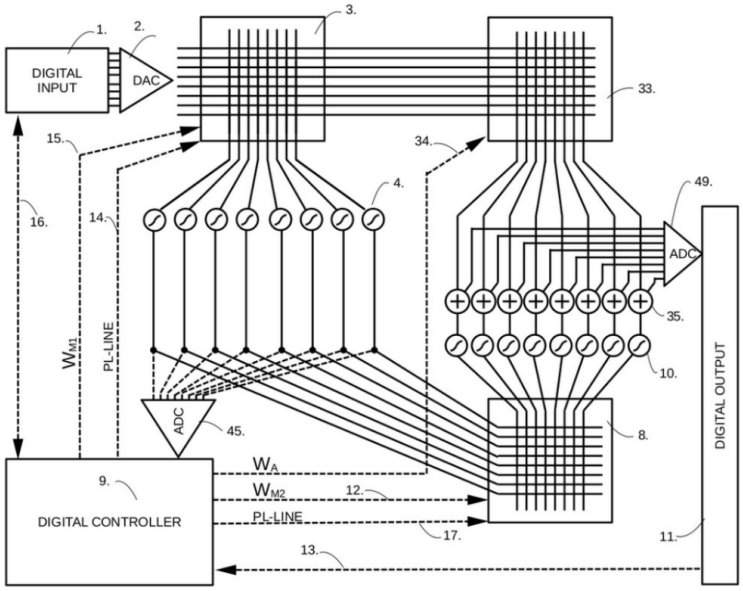


Fig 1

Figure 10: Inverted MAML Proprioceptive Feedback Architecture (Patent Application SE 2630397-4, Fig 1).
Digital input (1) is converted via DAC ladder (2) into analog signals driving both the primary uncompensated matrix Matrix 33 (33) and the first Vernier matrix Matrix 3 (3). The output of Matrix 3 passes through analog sigmoid activation (4) and is sampled by ADC (45) for the digital controller (9). The second Vernier matrix Matrix 8 (8) injects anti-phase compensation currents into summing junctions (35) before final transimpedance amplification and sampling by ADC (49) to produce the high-precision digital output (11).

11. Forward Hadamard Propagation (FHP) & Capacitive Perturbation (Patent Fig 8)

Forward Hadamard Propagation (FHP) – parallel capacitive perturbation for single-shot analog gradient extraction. (Works for physical unknown Analog Neural Networks).

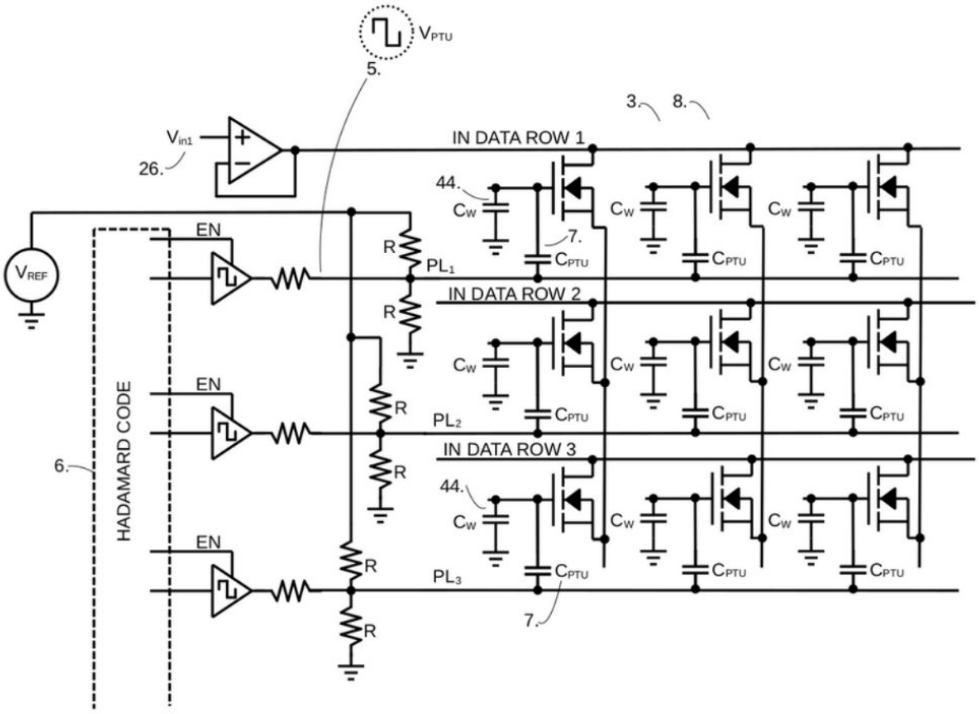


Fig 8

Physical Circuit Operation & Theoretical Breakthrough of FHP (SE 2630397-4 Fig 8):

1. Parallel Capacitive Perturbation (C_PTU, Ref 7): An AC perturbation square wave V_PTU (5) is modulated by 3-state Hadamard drivers (6) along perturbation lines PL_1..PL_n. Each crosspoint cell couples the carrier through dedicated perturbation capacitor C_PTU (7) onto the floating gate storage node C_W (44) of the 2T1C MOS triode transistor. Crucially, C_PTU perturbs the operating point dynamically without altering or leaking the static stored weight voltage.

2. Single-Shot Analog Gradient Extraction: In conventional analog tuning, extracting the gradient vector of N weights requires 2N sequential forward perturbations, creating massive latency and thermal drift. FHP applies mutually orthogonal Walsh/Hadamard carrier waves simultaneously across all rows. In a single analog measurement snapshot (t = 5.0 ms), the composite response at the summing amplifiers is projected onto the Hadamard basis in the digital domain (T120 SoC), isolating all partial derivatives in O(1) time.

3. Universal In-Situ Training for Arbitrary Analog Networks: FHP requires zero internal analytical models of the physical transistors, threshold voltages, parasitic capacitances, or thermal drift. It treats the physical Analog Neural Network as a true unknown black box, enabling robust closed-loop learning on physical silicon where classical backpropagation fails.

4. Complete Source Code & Hardware Repository: https://github.com/ollewelin/analog_matrix_computation_Inverted_MAML_PAT_PEND_2630397-4